

## CORPUS CHRISTI BUOY 42020, USA

WMO: 996190

Lat: 26.968N

Lon: 96.693W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (W06)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 47.9       | 51.2 | 33.3                          | 27.9 | 56.8 | 37.0 | 32.3 | 57.4 | 34.6                  | 59.9 | 32.8 | 59.4 | 21.5                  | 350  | 0.578 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 2.4                    | 86.1            | 78.4 | 85.5 | 78.7 | 85.2 | 78.6 | 82.2                | 83.8 | 81.2 | 83.7 | 80.4 | 83.5 | 10.8                 | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.7                            | 164.9 | 83.5 | 80.4 | 157.7 | 83.2 | 79.5 | 153.2 | 83.0 | 45.9          | 83.9 | 44.7 | 83.6 | 44.0 | 83.5 | 85.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 31.7              | 26.9 | 24.7 | 45.1                       | 86.7 | 3.9                | 1.0 | 42.3                                               | 87.5 | 40.1       | 88.0 | 37.9       | 88.6 | 35.1       | 89.4 |      |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |      |
|                   |      |      | WB                         | 41.7 | 81.8               | 2.9 | 1.6                                                | 39.7 | 82.9       | 38.0 | 83.9       | 36.4 | 84.7       | 34.3 | 85.9 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 74.5   | 64.6  | 64.6  | 66.9  | 71.7  | 77.1  | 81.7  | 83.0  | 83.5  | 82.1  | 78.5  | 72.8  | 67.1 |
|                                                                               | DBStd       | 8.04   | 5.63  | 5.62  | 4.33  | 3.21  | 2.62  | 1.43  | 1.26  | 1.30  | 2.10  | 3.40  | 4.74  | 5.77 |
|                                                                               | HDD50       | 5      | 2     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 230    | 74    | 65    | 32    | 3     | 0     | 0     | 0     | 0     | 0     | 9     | 47    |      |
|                                                                               | CDD50       | 8958   | 455   | 412   | 523   | 652   | 841   | 950   | 1024  | 1038  | 962   | 884   | 685   | 532  |
|                                                                               | CDD65       | 3709   | 63    | 55    | 90    | 205   | 376   | 500   | 559   | 573   | 512   | 419   | 244   | 113  |
|                                                                               | CDH74       | 33477  | 0     | 0     | 1     | 285   | 2594  | 5653  | 6820  | 7225  | 6035  | 3753  | 1026  | 85   |
|                                                                               | CDH80       | 9093   | 0     | 0     | 0     | 0     | 55    | 1412  | 2386  | 2784  | 1911  | 539   | 6     | 0    |
| Wind                                                                          | WSAvg       | 14.3   | 16.0  | 15.2  | 14.3  | 14.3  | 14.7  | 13.7  | 13.4  | 11.6  | 12.3  | 14.4  | 16.1  | 16.4 |
| Precipitation                                                                 | PrecAvg     | 34.7   | 2.2   | 1.9   | 2.2   | 2.0   | 2.6   | 3.2   | 2.4   | 2.6   | 6.5   | 4.3   | 2.6   | 2.2  |
|                                                                               | PrecMax     | 48.0   | 7.0   | 5.4   | 7.2   | 6.5   | 6.0   | 7.8   | 10.1  | 6.7   | 15.4  | 12.8  | 8.5   | 6.3  |
|                                                                               | PrecMin     | 20.1   | 0.3   | 0.2   | 0.3   | 0.1   | 0.1   | 0.4   | 0.4   | 0.4   | 1.5   | 0.2   | 0.6   | 0.7  |
|                                                                               | PrecStd     | 6.9    | 1.4   | 1.3   | 1.6   | 1.7   | 1.5   | 1.9   | 2.4   | 1.7   | 3.4   | 2.9   | 1.7   | 1.3  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 73.4  | 72.8  | 74.1  | 78.3  | 82.0  | 85.3  | 86.0  | 86.5  | 86.6  | 85.0  | 81.0  | 76.9 |
|                                                                               |             | MCWB   | 69.6  | 70.2  | 70.8  | 73.2  | 77.9  | 77.9  | 78.1  | 78.4  | 78.2  | 79.6  | 76.6  | 72.7 |
|                                                                               | 2%          | DB     | 72.9  | 72.1  | 73.1  | 77.4  | 81.3  | 84.4  | 85.4  | 86.3  | 85.9  | 84.0  | 79.8  | 75.9 |
|                                                                               |             | MCWB   | 69.2  | 69.5  | 70.2  | 73.9  | 77.8  | 78.0  | 78.1  | 78.5  | 78.4  | 78.3  | 74.6  | 71.8 |
|                                                                               | 5%          | DB     | 72.2  | 71.4  | 72.3  | 76.7  | 80.6  | 84.0  | 85.0  | 85.9  | 85.3  | 83.3  | 79.1  | 75.1 |
|                                                                               |             | MCWB   | 68.6  | 68.9  | 69.7  | 73.5  | 77.1  | 78.0  | 78.2  | 78.7  | 78.5  | 77.4  | 73.6  | 71.2 |
|                                                                               | 10%         | DB     | 71.4  | 70.8  | 71.5  | 75.9  | 80.2  | 83.5  | 84.5  | 85.4  | 84.9  | 82.7  | 78.3  | 74.2 |
|                                                                               |             | MCWB   | 67.8  | 68.3  | 69.0  | 73.0  | 76.6  | 78.1  | 78.2  | 78.8  | 78.1  | 76.7  | 72.7  | 70.6 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 71.5  | 71.3  | 72.4  | 76.9  | 81.1  | 82.4  | 83.4  | 83.4  | 82.5  | 81.9  | 79.7  | 75.3 |
|                                                                               |             | MCDB   | 72.5  | 71.6  | 73.2  | 77.2  | 81.2  | 83.4  | 83.5  | 84.3  | 85.0  | 83.5  | 79.8  | 75.6 |
|                                                                               | 2%          | WB     | 70.7  | 70.5  | 71.5  | 75.3  | 80.1  | 81.1  | 82.2  | 81.8  | 80.5  | 80.3  | 76.5  | 73.7 |
|                                                                               |             | MCDB   | 72.0  | 71.0  | 72.3  | 76.2  | 80.5  | 83.3  | 83.4  | 84.6  | 84.4  | 83.0  | 78.6  | 74.3 |
|                                                                               | 5%          | WB     | 69.7  | 69.9  | 71.0  | 74.8  | 78.9  | 80.2  | 81.0  | 80.6  | 79.8  | 79.3  | 75.1  | 72.7 |
|                                                                               |             | MCDB   | 71.4  | 70.6  | 71.9  | 75.8  | 79.7  | 83.0  | 83.4  | 84.4  | 84.1  | 82.6  | 77.6  | 73.8 |
|                                                                               | 10%         | WB     | 68.6  | 69.1  | 70.2  | 74.1  | 77.9  | 79.5  | 80.2  | 80.1  | 79.1  | 78.1  | 73.9  | 71.7 |
|                                                                               |             | MCDB   | 70.7  | 70.1  | 71.3  | 75.2  | 79.2  | 82.8  | 83.4  | 84.3  | 83.9  | 82.0  | 77.4  | 73.4 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 5.9   | 5.3   | 3.9   | 2.7   | 2.0   | 2.0   | 1.9   | 2.4   | 3.2   | 3.2   | 4.4   | 5.9  |
|                                                                               |             | MCDBR  | 5.0   | 4.2   | 3.2   | 2.4   | 2.3   | 2.5   | 2.0   | 2.5   | 3.0   | 3.0   | 4.1   | 5.3  |
|                                                                               |             | MCWBR  | 6.4   | 5.4   | 3.5   | 3.3   | 2.8   | 2.9   | 2.4   | 2.5   | 3.2   | 3.9   | 5.5   | 7.6  |
|                                                                               | 5% WB       | MCDBR  | 5.2   | 4.2   | 2.9   | 2.6   | 2.6   | 2.2   | 2.0   | 2.2   | 3.1   | 3.1   | 4.7   | 5.8  |
|                                                                               |             | MCWBR  | 6.9   | 5.4   | 3.6   | 3.2   | 3.0   | 3.1   | 3.1   | 2.9   | 3.7   | 4.5   | 6.7   | 7.2  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.376  | 0.387 | 0.397 | 0.452 | 0.484 | 0.453 | 0.471 | 0.454 | 0.443 | 0.401 | 0.384 | 0.373 |      |
|                                                                               | taud        | 2.479  | 2.447 | 2.420 | 2.258 | 2.199 | 2.364 | 2.321 | 2.377 | 2.363 | 2.483 | 2.490 | 2.499 |      |
|                                                                               | Ebn at Noon | 276    | 282   | 285   | 270   | 260   | 267   | 262   | 266   | 267   | 273   | 271   | 272   |      |
|                                                                               | Edh at Noon | 30     | 34    | 36    | 44    | 47    | 39    | 41    | 39    | 38    | 32    | 30    | 29    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 949    | 1160  | 1491  | 1790  | 2032  | 2177  | 2132  | 2041  | 1683  | 1478  | 1103  | 857   |      |
|                                                                               | RadStd      | 111    | 180   | 154   | 173   | 122   | 99    | 157   | 97    | 129   | 132   | 115   | 116   |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (23 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +189  |

Nomenclature: See separate page